Group 18

Pet Minder

ECS506U Software Engineering Group Project

Requirements Elicitation Report

2 Requirements

This chapter contains all requirements that have been decided upon with the customer. Pet Minder will adhere to the requirements listed in this chapter. The requirements have been prioritised using the MoSCoW method. Short explanation of this method including abbreviations can be found below.

Priority	Abbreviation	Explanation
Must have	M	These requirements are fundamental and must therefore be included in the product.
Should have	S	These requirements are important, but not as important as the must have requirements. Hence, they can be left out if there is no other choice.
Could have	C	These requirements are good to have in the product, but can easily be left out. These will only be implemented where there is time left or if they are east to implement.
Won't have	W	These requirements will not be implemented into the project but can be realised at a later version of the product.

2.1 Capability Requirements

2.1.1 General

ID	Requirement	Priority
URF 1.1	The system will allow a user to register as a Pet Owner, Pet Minder or both.	M
URF 1.2	The system will require users to provide a valid email address during registration.	M
URF 1.3	The system will require users to create a password during registration.	M
URF 1.4	The system will support two-factor authentication during login.	S
URF 1.5	The system will allow registered users to log into their account.	M
URF 1.6	System shall allow users to log out successfully.	M
URF 1.7	System will allow users to update their account information.	M
URF 1.8	The system will allow users to delete their account.	S
URF 1.9	The system will store the user's general location for matching purposes.	M
URF 1.10	The system shall restrict access to features based on user role.	M
URF 1.11	The system shall allow users to switch between Pet Minder and Pet Owner role.	M
URF 1.12	The system shall allow users to review one another (pet owner – pet minder) rating from 1-5	M
URF 1.13	The system shall allow users to leave a written review	S
URF 1.14	The system will calculate and display the average rating of a Pet Minder.	M

2.1.2 Pet Owner

2.1.2.1 Pet Profiles

ID	Requirement	Priority
URF 2.1	The system shall allow a Pet Owner to create one or more pet profiles.	M
URF 2.2	Each pet profile shall include a name.	M
URF 2.3	Each pet profile shall include the pet type (i.e. dog, cat, lizard…).	M

URF 2.4	Each pet profile should include the pet's age.	S
URF 2.5	The system shall allow a Pet Owner to add medical information to a pet profile.	M
URF 2.6	The system will allow a Pet Owner to add dietary requirements to a pet profile.	S
URF 2.7	The system will allow a Pet Owner to edit an existing pet profile.	M
URF 2.8	The system will allow a Pet Owner to delete a pet profile.	M

2.1.2.2 Search & Scheduling

ID	Requirement	Priority
URF 2.9	The system will allow a Pet Owner to search for Pet Minders by location.	M
URF 2.10	The system will allow a Pet Owner to filter search results by availability.	M
URF 2.11	The system will allow a Pet Owner to filter search results by rating.	S
URF 2.12	The system shall display a list of matching Pet Minders.	M
URF 2.13	The system shall allow a Pet Owner to view a Pet Minder's profile.	M
URF 2.14	The system will allow a Pet Owner to request a booking.	M
URF 2.15	The system shall require the Pet Owner to select date and time when requesting a booking.	M
URF 2.16	The system shall allow a Pet Owner to attach care instructions to a booking.	M
URF 2.17	The system shall allow a Pet Owner to cancel a booking up to 48 hours before the scheduled time (depending on task as something like dog walking this won't apply).	M
URF 2.18	The system shall allow a Pet Owner to reschedule a booking before confirmation by the Pet Minder.	S
URF 2.19	The system shall allow a Pet Owner to cancel a job minimum 3 days before it is scheduled.	M

2.1.2.3 Monitoring

ID	Requirement	Priority
URF 2.20	The system shall allow a Pet Owner to view live activity tracking during a session.	M
URF 2.21	The system shall allow a Pet Owner to view completed activity logs.	S

2.1.3 Pet Minder

ID	Requirement	Priority
URF 3.1	The system shall allow a Pet Minder to create and manage their professional profile, including personal details, qualifications, experience and service descriptions.	M
URF 3.2	The system shall allow a Pet Minder to specify supported pet types.	M
URF 3.3	The system shall allow a Pet Minder to set and update their service pricing.	M
URF 3.4	The system shall allow a Pet Minder to set and update their availability schedule.	M
URF 3.5	The system shall allow the Pet Minder to view incoming booking requests.	M
URF 3.6	The system shall allow the Pet Minder to accept or decline a booking request.	M
URF 3.7	The system shall allow a Pet Minder to cancel a confirmed booking up to 48 hours before the scheduled time.	M

URF 3.8	The system shall allow a Pet Minder to view booking details, including pet information and owner instructions.	M
URF 3.9	The system shall allow a Pet Minder to start a pet minding session.	M
URF 3.10	The system shall allow a Pet Minder to terminate a pet minding session.	M
URF 3.11	The system shall allow a Pet Minder to record and submit a session activity log for a completed booking.	M
URF 3.12	The system shall allow a Pet Minder to update live activity tracking during a session.	M

2.1.4 Administrator

ID	Requirement	Priority
URF 4.1	The system shall allow an Administrator to view all registered users.	M
URF 4.2	The system shall allow an Administrator to suspend or deactivate user accounts.	M
URF 4.3	The system shall allow an Administrator to verify Pet Minder profiles before they are activated.	M
URF 4.4	The system shall allow an Administrator to resolve booking disputes between users.	M
URF 4.5	{The system shall allow an Administrator to moderate and remove inappropriate reviews	M
URF 4.6	The system shall log all administrative actions with accurate timestamps	M

2.2 Constraint Requirements

2.2.1 Privacy & Security

ID	Requirement	Priority
URC 1.1	Sensitive data must be encrypted while at rest within database	M
URC 1.2	Administrative actions are logged with timestamps and user identifiers.	M
URC 1.3	User sessions automatically expire after 15 minutes of inactivity.	S
URC 1.4	Explicit consent of user before collecting, storing or displaying live location for tracking activity is necessary	M
URC 1.5	Users will be locked out of their account if they're not able to login after 5 consecutive times	S

2.2.2 Usability

ID	Requirement	Priority
URC 2.1	User can perform an implemented use case within 15 minutes (without instructions)	M
URC 2.2	User can perform an implemented use case within 10 minutes (without instructions)	S
URC 2.3	The booking process must be completed under 5 steps	S
URC 2.4	The UI should provide an immediate response to all user triggered actions	M
URC 2.5	The UI should remain consistent across both mobile and desktop devices	M
URC 2.6	All non-text content in UI of application should have alternative texts	C

2.2.3 Environment

ID	Requirement	Priority
URC 3.1	Pet Minder functions correctly on the latest stable versions of Google Chrome, Microsoft Edge, Mozilla Firefox and Apple Safari.	M
URC 3.2	Pet Minder shall be available in English	M
URC 3.3	Pet Minder shall be using Next.js (React) framework	M
URC 3.4	Pet Minder shall be using Supabase for backend services and deployed using Vercel cloud hosting	M
URC 3.5	Pet Minder shall provide users option to switch between languages	C

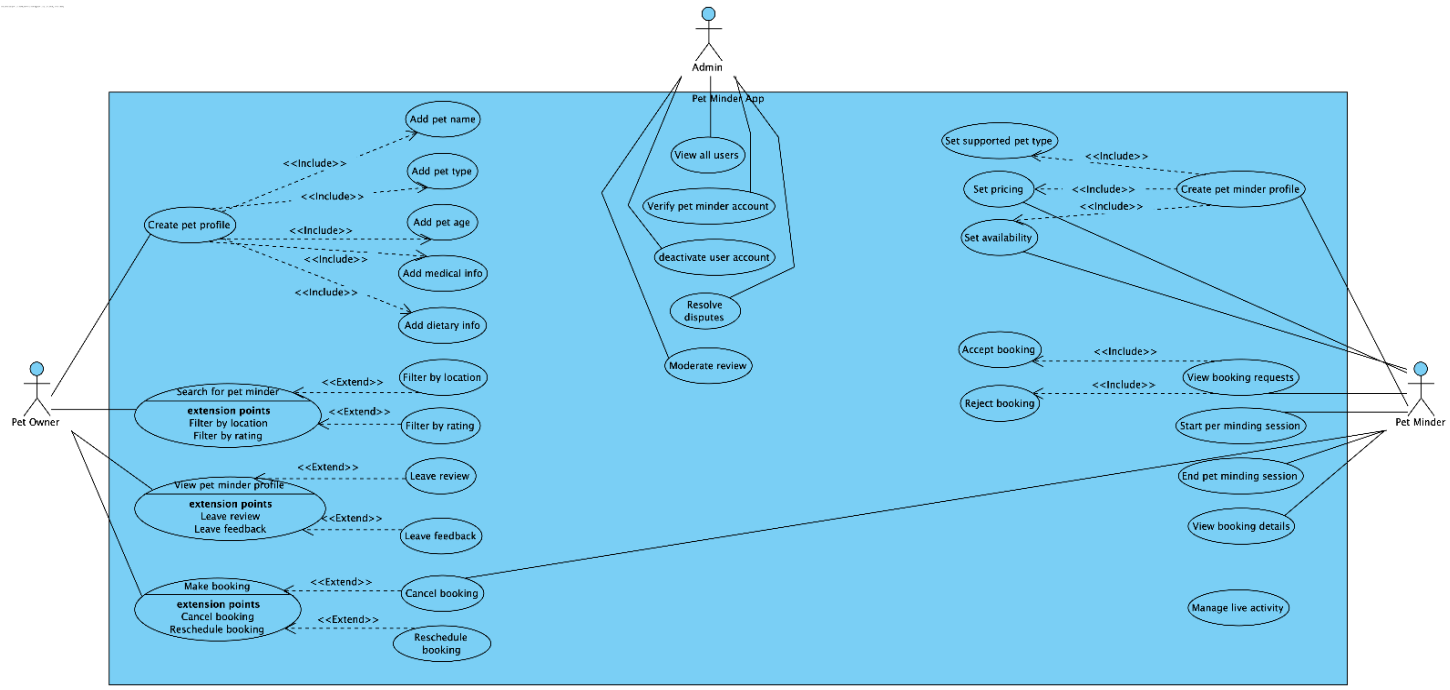
2.2.4 Performance

ID	Requirement	Priority
URC 4.1	99% of user requests complete within 3 seconds under normal load	S
URC 4.2	Live tracking updates occur with a delay no more than 5 seconds	S
URC 4.3	Search results of pet minders are returned within 3 seconds	C

2.2.3 Reliability

ID	Requirement	Priority
URC 5.1	Pet Minder shall maintain 99% availability in a 24-hour period time	S
URC 5.2	Pet Minder restores critical functionality within 2 hours after a failure	S
URC 5.3	Critical data should be backed up daily	M

1. Use-Case Diagram



2. Use Case Description

Use Case 1

Name

Make Booking

Brief Description

The pet owner can search for a pet minder that provides a walking services within a set location radius and can schedule if for a date and time of their convenience.

Actors

- Pet Owner (Primary)
- Pet Minder (Secondary)

Preconditions

- The pet owner is logged in
- The pet owner has set up pet profiles for all the pets being walked
- The pet owner has location services enabled

Basic Flow

1. The pet owner chooses 'book a service' from the dashboard
2. The system shows all the pet profiles
3. The pet owner needs to select the pets being walked (the system prevents selecting more than 4)
4. The system asks for the date, time and length of the walk
5. The pet owner provides the desired date, time and duration
6. The system filters pet minders according to availability and capacity to walk those specific pets
7. The system displays the minders that match the filters along with their price
8. The pet owner selects a pet Minder and clicks 'request booking'
9. The system sends notification to the pet minder with the request
10. The pet minder checks the request and clicks 'accept'
11. The system puts the walk in the pet minders calendar and sends a reminder to the pet owner

Alternate Flows

Incompatible Pets Selected

3a. The pet owner choses pets with conflicting behaviours
3b. The system displays a warning 'Pets have conflicting behaviours. Recommended to go on separate walks.'
3c. The pet owner overrides suggestion and proceeds with pets walking together or changes their selection

No minders available for the selected group size

6a. The pet owner has selected a valid number of pets but there isn't anyone available in the selected area who would walk that number of pets
6b. The system displays 'No minders available for this group size'
6c. The system displays a suggestion 'Try selecting fewer pets, increasing search radius or expanding your availability'
6d. Return to step 3

Minder Declines

10a. The pet minder views request and clicks 'decline'
10b. The system notifies Pet Owner and returns them to step 7.

Post Conditions

- A confirmed booking created in the database

- Pet minder is no longer available for that time
- The booking appears in both users' accounts

Use Case 2

Name

View booking requests

Brief Description

A pet minder can view incoming booking requests from Pet Owners, review the details of the pets and the requested time slot, and choose to accept or decline the request. The system updates availability and notifies the pet owner accordingly.

Actors

- Pet Minder (Primary)
- Pet Owner (Secondary)

Preconditions

- The pet minder is logged in.
- The pet minder has an active profile with availability set.
- A booking request has been submitted by a pet owner.

Basic Flow

1. The system sends a notification to the Pet Minder about a new booking request.
2. The pet minder navigates to the "Requests" section from the dashboard.
3. The system displays all pending booking requests.
4. The pet minder selects a specific request to view details.
5. The system displays the pet(s) involved, requested date, time, duration and any notes/care instructions.
6. The pet minder reviews the information and clicks "Accept".
7. The system checks for scheduling conflicts.
8. If no conflict exists, the system confirms the booking, updates the Pet Minder's availability, and sends a confirmation to the Pet Owner.

Alternate Flows

Decline Request

6a. Instead of accepting, the Pet minder clicks "Decline".

6b. The system prompts for an optional reason (e.g. unavailable etc).

6c. The Pet minder submits the decline.

6d. The system notifies the Pet owner that the request was declined.

Scheduling Conflict

7a. The system detects an overlapping booking.

7b. The system displays "You already have a booking at this time."

7c. The pet minder chooses either:

- Decline --> Follow steps 6a – 6d.
- Return to "Requests" without taking action.

Post Conditions

- A confirmed booking is stored in the system (if accepted).

- The pet minders availability is updated.
- The pet owner receives a notification of acceptance or decline.
- The booking appears in both users' accounts (if accepted).

3. Risk Assessment

Risk	Likelihood	Severity	Impact	Preventative / Mitigating Actions
Poor time management	High	High	Unfinished product	Evaluate progress frequently and adjust project plan if necessary
Missing member(s) due to health issues	High	Medium	Unfinished/ low quality product	Have frequent meetings and communication to make sure all members aware of each other's progress.
Data Breach	Medium	High	Loss of user trust, legal consequences, reputational damage	Implement secure authentication (2FA), encrypt sensitive data, limit access to stored information
GPS Tracking Failure	Medium	Medium	Reduced trust from users and inaccurate activity monitoring	Use reliable software, test tracking across multiple devices, include backup mechanisms
GDPR or Legal Non-Compliance	Low	High	Legal penalties and forced system redesign	Ensure clear use of consent for data collection, follow data minimisation principles, review GDPR requirements carefully
Unreliable/Fraudulent Pet Minder	Medium	High	Harm to pets, negative reviews, reputational damage	Implement identity verification, background checks if applicable, two-sided review system, reporting mechanisms
Injury to pet	Low	High	Harm to pet, legal claims, reputational damage	Require minders to agree to safety guidelines, allow emergency contact information, include clear care instructions
Injury to minder	Low	Medium	Legal issues and service disruption	Include safety guidelines, allow minders to decline high-risk pets, provide clear pet behaviour details
System downtime/Failure	Medium	High	Users unable to book or track services, reduce reliability	Use reliable cloud hosting, maintain backups, monitor system performance
Miscommunication between owner and minder	Medium	Medium	Poor service delivery, negative reviews	Provide in-app messaging, booking confirmations, clear instruction fields

Low user adoption/Market Competition	Medium	Medium	Limited platform growth	Focus on user friendly design, competitive features, effective promotion strategy
Adding too many features	Medium	Medium	Delays and incompletes core functionality	Define requirements early, prioritise essential features, review changes as a team